Stony Brook Institute at Anhui University

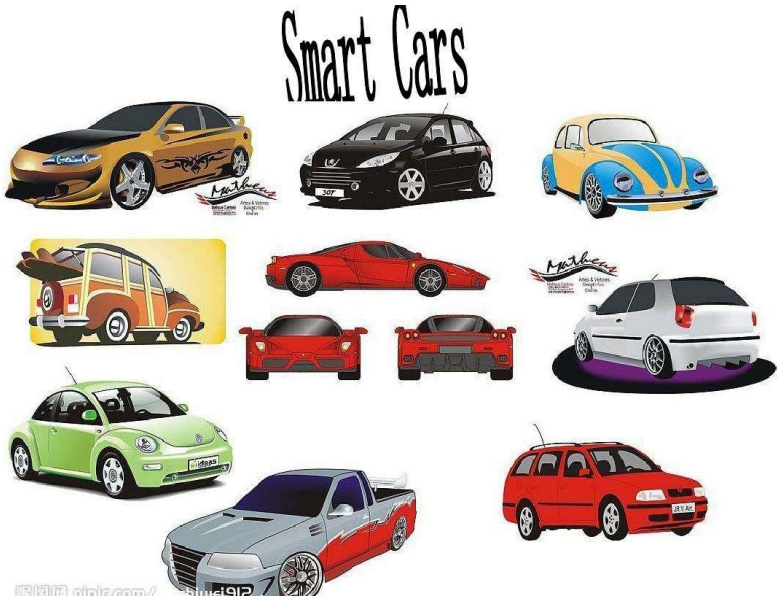
安徽大学纽约石溪学院

Chapter 5
Basic Concepts of Statistics

Introudution to Statistics

Statistics provides methods for：

- Data collection；
- Data summarizing and interpreting;
- Conclusions.



Introudution to Statistics

- Statistical thoughts and methods are not only useful but indeed often indispensable in understanding the world around us.
- Statistics provide ways of gaining new insights into the behavior of many phenomena in the chosen field of specialization in engineering or science.

5.1 Population and random samples

- Population and samples
- Independent and identically distributed random samples (i.i.d.)

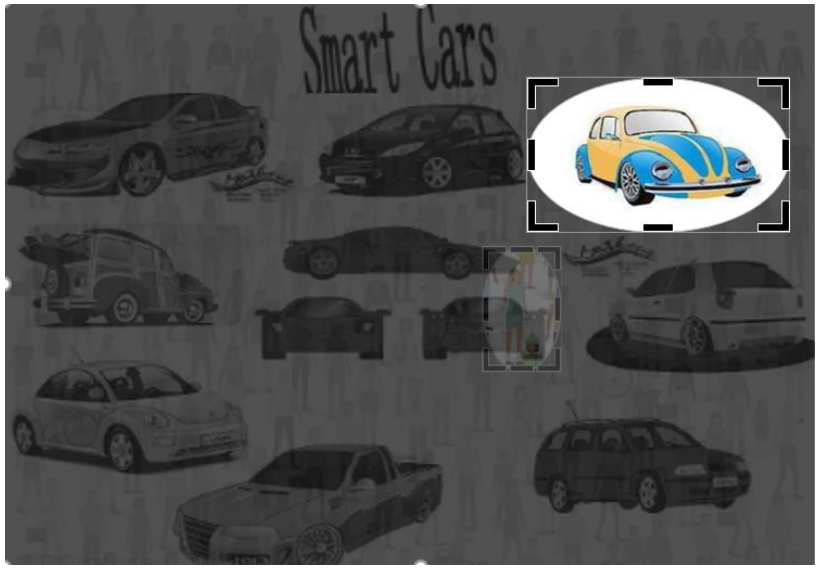
Some basic concepts

- **Populations**

The vast collection of all potential observations of a particular type;

- **Individuals**

Every object included in the population.



Some basic concepts

Population's property:

- Population is a r.v. with a certain distribution F(x). X,Y,....are often used to represent population, that means population

r.v. X ~ F(x)

- we use the distribution's name to call the population, such as

  a. normal population X, this means r.v. X~N(μ, σ^2)

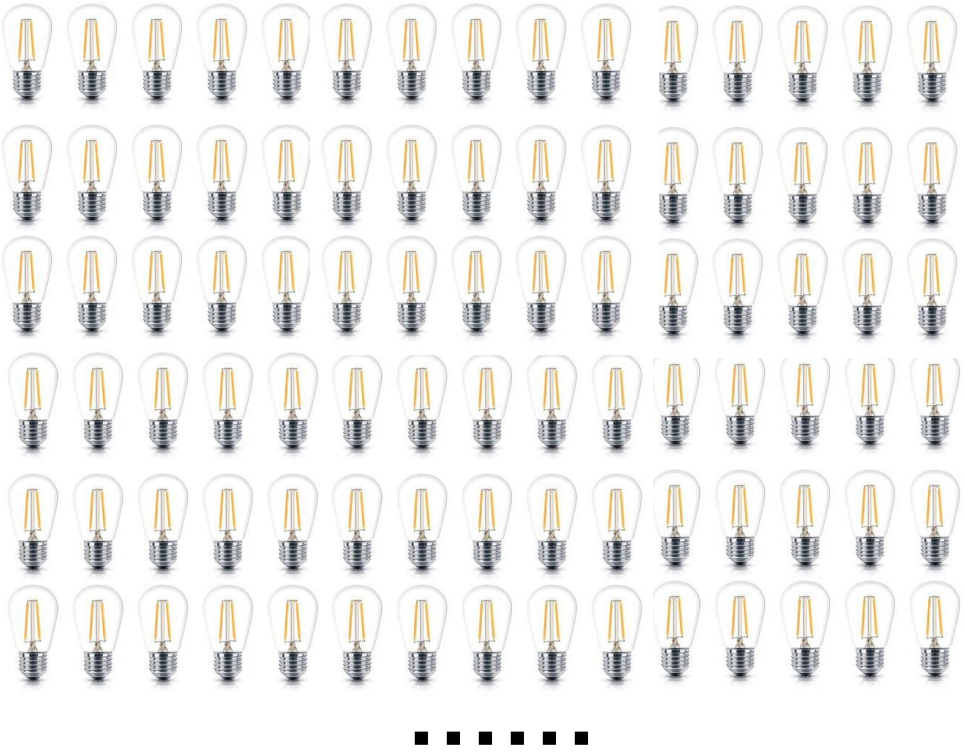
  b. binominal population X, this means r.v. X~B(n,p)

If a population is infinite, it is impractical to observe it entirely. Thus, it is necessary to use samples of the population to infer some quantities that characterize the entire population.

Basic conception of sampling

- **Sample**

The data set that is acquired through the process of the population observation.



Independent and identically distributed random samples

- **Simple random sample of size *n*:**

The random variables $\boldsymbol{X_1, X_2, \ldots, X_n}$ are a random sample with size n, if:

1) $\boldsymbol{X_1, X_2, \ldots, X_n}$ are independent, and
2) every $\boldsymbol{X_i}$ has the same distribution.

In other words, $\boldsymbol{X_1, X_2, \ldots, X_n}$ are independent and identically distributed(i.i.d.), and have the same distribution as population X, that means

$$\boldsymbol{X_1, X_2, \ldots, X_n} \quad \text{i.i.d.} \sim X$$

Joint distribution of samples

Suppose that the population X has the cdf $F(x)$ or pdf $f(x)$, and samples $\boldsymbol{X_1, X_2, \ldots, X_n}$ i.i.d. ~ X , so

- the joint pdf of samples is

$$\boldsymbol{f_n(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} f(x_i)}$$

- the joint cdf of samples is

$$\boldsymbol{F_n(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} F(x_i)}$$

5.2 Statistic and Sample Distribution

- The information comes from the sample are too diverse to draw useful conclusion. So we'll summarize the sample information through statistics for use.
- Statistical inference is directed toward drawing some type of conclusion about one or more population quantities, such as the mean μ and variance σ^2.
- The population quantities are called parameters.
- **A function of one or more samples that is not dependent on any unknown parameter is called a Statistic.**

5.2 Statistic and Sample Distribution

- Statistic
- Common statistic
- Common sample distribution

Statistic

Definition:

Statistic is the function of samples that is **not** dependent on any **unknown parameter**, it is denoted as

$$T=g(\boldsymbol{X}_1,\boldsymbol{X}_2,\ldots,\boldsymbol{X}_n)$$

- Statistic T is uncorrelation to any unknown parameters;
- Statistic T is a r.v. and has a probability distribution F(t).

Statistic

Q :If X_1, X_2, X_3 are the samples comes from Normal population $N(\mu, \sigma^2), \mu$ is known, σ^2 is unknown. Which are statistics in the following expression?

$$T_1 = X_1,$$

$$T_2 = X_1 + X_2 e^{X_3},$$

$$T_3 = \frac{1}{3}(X_1 + X_2 + X_3),$$

$$T_4 = \max(X_1, X_2, X_3),$$

$$T_5 = X_1 + X_2 - 2\mu,$$

$$T_6 = \frac{1}{\sigma^2}(X_1^2 + X_2^2 + X_3^2).$$

Common statistic

- Sample mean:

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- Sample variance:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right).$$

- Sample standard deviation:

$$S = \sqrt{S^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}\ ;$$

Common Statistic

- kth Sample (origin) moments:

$$A_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k \ , k = 1, 2, \cdots;$$

- k-th Sample central moments:

$$\boldsymbol{B_k = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^k \ , k = 2, 3, \cdots;}$$

- **Theorem 5.1** Suppose the *k*th sample moment of X exists, then $if\ n \to \infty,$

$$\frac{1}{n}\sum_{i=1}^{n} X_i^k \xrightarrow{P} E(X^k),\ k = 1, 2, \cdots.$$

The property of sample mean and variance

Theorem 5.2: Suppose population X satisfied,

$$E(X)=\mu, Var(X)=\sigma^2, \quad \text{then :}$$

$$(1)\ E(\overline{X})=\mu; \qquad (2)\ Var(\overline{X})=\tfrac{1}{n}\sigma^2;$$

$$(3)\ E(S^2)=\sigma^2;$$

$$E(X)=E(X_i)=\mu, \quad Var(X)=Var(X_i)=\sigma^2$$

$$(1)\ E(\overline{X})=E(\tfrac{1}{n}\sum_{i=1}^{n}X_i)=\tfrac{1}{n}\sum_{i=1}^{n}E(X_i)=\tfrac{1}{n}\sum_{i=1}^{n}\mu=\mu$$

$$(2)\ \mathrm{Var}(\overline{X})=Var(\tfrac{1}{n}\sum_{i=1}^{n}X_i)=\tfrac{1}{n^2}\sum_{i=1}^{n}Var(X_i)=\tfrac{1}{n^2}\sum_{i=1}^{n}\sigma^2=\tfrac{1}{n}\sigma^2$$

$$(3)\ E(S^2) = E[\tfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2] = \tfrac{1}{n-1}E(\sum_{i=1}^{n}X_i^2 - n\overline{X}^2)$$

$$= \tfrac{1}{n-1}[\sum_{i=1}^{n}E(X_i^2) - nE(\overline{X}^2)] = \tfrac{1}{n-1}[nE(X_1^2) - nE(\overline{X}^2)]$$

$$= \tfrac{n}{n-1}[Var(X_1) + (EX_1)^2 - (Var(\overline{X}) + (E\overline{X})^2)]$$

$$= \tfrac{n}{n-1}[(\sigma^2 + \mu^2) - (\tfrac{1}{n}\sigma^2 + \mu^2)] = \tfrac{n}{n-1}\cdot\tfrac{n-1}{n}\sigma^2$$

$$= \sigma^2$$

The property of sample mean

Theorem 5.3: Suppose sample $X_1, \ldots, X_n$ are i.i.d. with the mean μ and variance σ^2, t>0 is a constant, then by Chebyshev inequality, the sample mean $\bar{X}$ satisfied:

$$P(|\bar{X} - \mu| \geq t) \leq \frac{\sigma^2}{nt^2}$$

The property of sample mean

Example: Determine the required number of observation

Suppose n random samples are taken from a distribution with an unknown mean μ *and variance* $\sigma^2 = 4$.

Find out how large the sample size must be so that there at least a probability of 0.99 that $|\bar{X} - \mu|$ will be less than1.

Solution: For samples with size n, by Chebyshev inequality,

$$P(|\bar{X} - \mu| \geq 1) \leq \frac{\sigma^2}{n} = \frac{4}{n}.$$

Since n must be chosen so that

$$P(|\bar{X} - \mu| < 1) = 1 - P(|\bar{X} - \mu| \geq 1) \geq 1 - \frac{4}{n} \geq 0.99$$

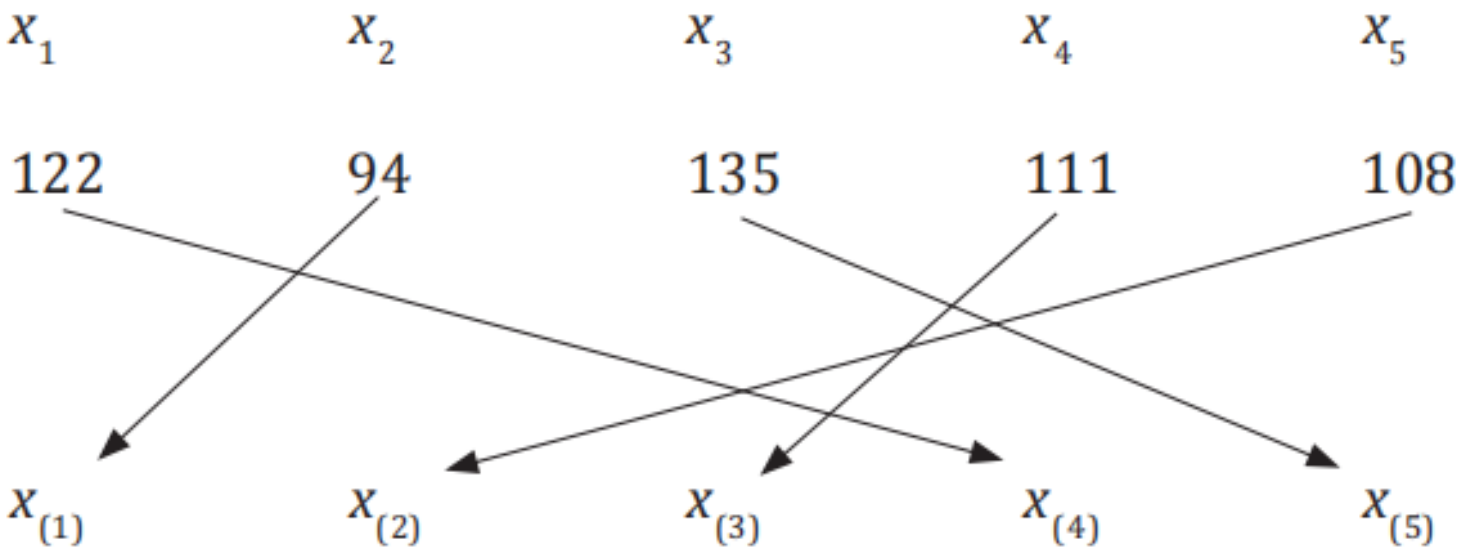
$$\Longrightarrow \quad \frac{4}{n} < 0.01 \Longrightarrow \quad n \geq 400.$$

Order statistic

➢ Order statistic:

Rearrange the elements $\boldsymbol{x_1, x_2, \ldots, x_n}$ of the random sample $\boldsymbol{X_1, X_2, \ldots, X_n}$ in an increasing order $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ to yield r.v. $\boldsymbol{X_{(1)}, \ldots, X_{(n)}}$.

Example:



$\boldsymbol{X_{(1)}, \ldots, X_{(n)}}$ are order statistic.

Order statistic

So we know

$$X_{(1)} = \min(X_1, X_2, \cdots, X_n)$$

$$X_{(n)} = \max(X_1, X_2, \cdots, X_n)$$

- $\boldsymbol{X_{(1)}, \ldots, X_{(n)}}$ are also random variables and they are often not independent .

Order statistic

If $F(x)$ and $p(x)$ are the cdf and pdf of population X, $(X_{(1)}, X_{(2)}, \cdots, X_{(n)})$ is the order statistic, then

(1) The pdf of $X_{(1)}$ is

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x)$$

(2) The pdf of $X_{(n)}$ is

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} f(x)$$

Example: Suppose population X~U[0, θ], $(X_1, \ldots X_n)$ are samples. What is the pdf of $X_{(1)}$, $X_{(n)}$?

Solution: The pdf and cdf of X are

$$f(x) = \begin{cases} \dfrac{1}{\theta}, & 0 \le x \le \theta \\ 0, & else. \end{cases} \qquad F(x) = \begin{cases} 0, & x < 0 \\ \dfrac{x}{\theta}, & 0 \le x \le \theta \\ 1, & x > \theta \end{cases}$$

We get $X_{(1)}$'s pdf is

$$f_{X_{(1)}}(x)=\begin{cases}\dfrac{n}{\theta}(1-\dfrac{x}{\theta})^{n-1}, & 0\le x\le\theta\\ 0, & else\end{cases}$$

and $X_{(n)}$'s pdf is

$$f_{X_{(n)}}(x)=\begin{cases}\dfrac{n}{\theta^{n}}x^{n-1}, & 0\le x\le\theta\\ 0, & else\end{cases}$$